

7(a)	$V / 9.0 = 1800 / (1800 + 1200)$	C1
	$V = 5.4 \text{ V}$	A1
	or	
	$I = 9.0 / (1800 + 1200) = 3.0 \times 10^{-3} \text{ (A)}$ $V = 3.0 \times 10^{-3} \times 1800$	(C1)
	$= 5.4 \text{ V}$	(A1)
7(b)(i)	$L / 1.2 = 5.4 / 9.0$ or $XZ / 1.2 = 5.4 / 9.0$	C1
	$L = 0.72 \text{ m}$	A1
	or	
	$L / 1.2 = 1800 / (1800 + 1200)$ or $XZ / 1.2 = 1.8 / (1.8 + 1.2)$	(C1)
	$L = 0.72 \text{ m}$	(A1)
7(b)(ii)	(intensity) increase	B1
	(power) increase	B1
	(length XZ) decrease	B1